Performance Assessment | D207 Exploratory Data Analysis

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A1. Question for Analysis

Do variables like gender, Yearly equivalent time, streaming TV, and having device protection have an impact on turnover, allowing us to better analyze the behavior of customers and minimize turnover and churn rate?

A2. Benefit for Analysis.

Organizational stakeholders will benefit from knowing, with some degree of certainty, which consumers are more likely to churn and which won't. On the dataset, the Churn feature will be used to determine customer churn, and the remaining features will be used to determine which variables are most predictive of customer churn. Knowing which customer features the customer values the most, is a result of analysis conducted, stakeholders will have their questions answered and will be able to make better business decisions.

A3. Data Identification.

We will examine all 10 features from the dataset. This dependent variable 'Churn,' which is binary categorical with only two values, 'Yes' or 'No,' will be more significant to our judgment process. Variables such as "Gender," "Multiple", "Device Protection" and "Streaming TV" will be analyzed to see if these features are dependent on "Churn" or not.

According to the data dictionary given by WGU, "Gender" refers to a customer's sex, such as "Male" or "Female." "Multiple" refers to whether a customer has multiple lines, or not. "Device Protection" represents whether the customer has a device protection add-on (yes), "Streaming TV" represents whether the customer has an ability to stream TV (yes/no) and "Multiple" represents whether the customer has multiple lines (yes/no).

B1. Code.

This data set will be analyzed using the CRI-Square test approach. Code is written in Python. Code is listed below

B2. Outputs and Results

```
import numpy as np
import matplotlib as mp
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd

%matplotlib inline
from sklearn import preprocessing
from sklearn.preprocessing import LabelEncoder
from scipy.stats import chi2_contingency

df = pd.read_csv('churn_clean.csv', usecols = ['Churn', 'Gender', 'Multiple',
'StreamingTV', 'DeviceProtection'])  #df is use to read the clean dataset. Only variables that needed to be analyzed are called upon.

df.dtypes
df.isnull().sum()  #checking for missing values
df['Gender'] = df.Gender.replace(to_dict{'Female', 'Male', 'Prefer not to answer'},
                                 inplace=True)
df['Churn'] = df.Churn.replace(to_dict{'No', 'Yes'},
                               inplace=True)
df['Multiple'] = df.Multiple.replace(to_dict{'No', 'Yes'},
                                     inplace=True)
df['StreamingTV'] = df.StreamingTV.replace(to_dict{'No', 'Yes'},
                                           inplace=True)
df['DeviceProtection'] = df.DeviceProtection.replace(to_dict{'No', 'Yes'},
                                                     inplace=True)
#recode the features
df.head(10)
```

```
Churn  Gender  Multiple  StreamingTV  DeviceProtection
0    No  Female        No          Yes                 No
1    No    Male       Yes          Yes                 No
2   Yes  Female       Yes           No                 No
3    No  Female        No           No                 No
4   Yes    Male       Yes          Yes                 No
5    No    Male        No          Yes                 No
6    No    Male        No           No                 No
7    No  Female        No          Yes                 No
8   Yes  Female       Yes          Yes                 No
9    No  Female       Yes          Yes                 No
```

```
df['Churn'].value_counts()
df['Gender'].value_counts()
df['Multiple'].value_counts()
df['StreamingTV'].value_counts()
df['DeviceProtection'].value_counts()
```

```
Churn
No    5174
Yes   1869
Name: count, dtype: int64

Gender
Female                 2541
Male                   2455
Prefer not to answer      2
Name: count, dtype: int64

Multiple
Yes    5106
No     1937
Name: count, dtype: int64

StreamingTV
No     3707
Yes    3336
Name: count, dtype: int64

DeviceProtection
No     3704
Yes    3339
Name: count, dtype: int64
```

```
df['Churn'].sort_values(ascending=False).value_counts(normalize=True)
df['Gender'].sort_values(ascending=False).value_counts(normalize=True)
df['Multiple'].sort_values(ascending=False).value_counts(normalize=True)
df['StreamingTV'].sort_values(ascending=False).value_counts(normalize=True)
df['DeviceProtection'].sort_values(ascending=False).value_counts(normalize=True)
```

```
Churn
No    0.73463
Yes   0.26537
Name: proportion of churn, dtype: int64

Gender
Female                 0.360966
Male                   0.348722
Prefer not to answer   0.000284
Name: proportion of gender, dtype: int64

Multiple
Yes    0.725004
No     0.274996
Name: proportion of multiple, dtype: int64

StreamingTV
No     0.526351
Yes    0.473649
Name: proportion of streamingtv, dtype: int64

DeviceProtection
No     0.525932
Yes    0.474068
Name: proportion of deviceprotection, dtype: int64
```

```
df.shape
df.dtypes
df.describe()
```

```
(7043, 5)

Churn              object
Gender             object
Multiple           object
StreamingTV        object
DeviceProtection   object
dtype: object
```

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